

Source Attribution of Watermarked Text

Han Li (hl2595) · Kangbo Hao (kh873) · Ruoxuan Cao (rc986)
M.Eng. Design Project · ECE, Cornell · Advisor: Prof. Vikram Krishnamurthy · ECE 5995/5996 · 2025–2026

Motivation & Problem

Multiple LLMs embed cryptographic watermarks for provenance. When an observer receives text, which model produced it? Two attribution paradigms exist:

Watermark (Active)	Fingerprint (Passive)
- Per-model KGW key - z-test each key - Strong on clean text	- TF-IDF n-gram features 4-way classifier - No key needed

Research Gaps

- Does watermarking compress inter-model distributional differences?
- Head-to-head watermark vs. fingerprint attribution under attack?
- Can sentence-level watermarks (SimMark) complement token-level (KGW)?

Method — KGW Watermarking

Kirchenbauer et al., ICML 2023. Token-level green/red partition with logit bias. Detection via z-test on green token count.

$$Z = (|s|_G - \mathbb{E}T) / \sqrt{(\mathbb{E}(1-\mathbb{E})T)}$$

Parameters:
 $\mathbb{E} = 0.25$ · $\mathbb{E} = 2.0$ · z-threshold = 3.0

Models:
LLaMA-3.2-1B, LLaMA-3.2-3B, Gemma-2B, Phi-2

Attacks:
Synonym sub. 10–70%, Ins/Del 10–30%, Reorder, Neural (Qwen-3)

F1 Distributional Compression

Watermarking shrinks inter-model differences to 1/3 of clean baseline (Gemma-2B vs Phi-2).

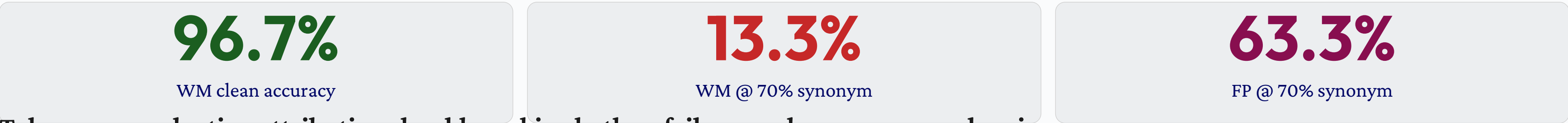
Setting	Wt.Diff	Ent.Diff
Indep. WM keys	0.24	0.76
No WM (clean)	0.76	2.30
Shared green list	1.56	4.48

Green-list bias dominates each model's natural signature — a direct headwind for passive fingerprint attribution.

F2 Robustness Crossover

Two paradigms fail in non-overlapping regimes. Watermark: cliff-like collapse. Fingerprint: graceful degradation. Lines cross at 50–70% synonym substitution.

Synonym %	Watermark	Fingerprint	Winner
0% (clean)	96.7%	76.7%	Watermark
10–30%	93–97%	70–77%	Watermark
50%	73.3%	66.7%	Watermark
70%	13.3%	63.3%	Fingerprint



Takeaway: production attribution should combine both — failure modes are non-overlapping.

F3 Hybrid KGW + SimMark

SimMark (EMNLP 2025): sentence-level cosine similarity watermark. Combined with KGW via rejection sampling. Detection: OR logic — either positive watermarked.

SimMark vs KGW — Moderate Paraphrase (5 rounds)

Metric	SimMark	KGW
Survival Rate	80%	20%
z-score Decay	34%	75%

Hybrid Robustness (OR-logic detection)

Scenario	KGW	SimMark	Hybrid OR
Short, no atk	66.7%	33.3%	66.7%
Poetry, no atk	66.7%	66.7%	100% $\square$
Long, mod. atk	0%	100%	100%
Any, aggressive	0%	0%	0%

KGW Rescues	SimMark Rescues
- Short text (≤ 3 sent.): SimMark lacks stats - Haiku: KGW z=5.89, SimMark z=−0.58	- Moderate rewrite: semantics preserved - Long: SimMark z=3.87, KGW z=1.66

F4 Multi-LLM Chain Paraphrase

17 experiments: Llama-3.2-3B Qwen-3 paraphraser. KGW watermarks do not survive neural paraphrase.

	17 chain experiments	6% survival (1/17)	5.07 avg z-decay
Mode	Orig z	After z	Survived
Standard	5.73	−1.16	No
Creative	5.73	0.77	No
Structure	5.73	0.05	No
Summary-exp. Neural paraphrase is far more destructive than symbolic synonym substitution.	5.73	−0.20	No

Conclusions

- Watermarking compresses distributions — makes different models look more alike.
- Watermark & fingerprint: non-overlapping failure modes (crossover at 50–70%).
- KGW + SimMark are complementary — Hybrid OR achieves 100% on poetry.
- Neural paraphrase defeats KGW alone — only 1/17 survived.
- Aggressive attack remains an open challenge for all current methods.

Future Work

- Hybrid attribution classifier: KGW z-scores + TF-IDF features in one model
- Neural paraphrase attacks: extend with GPT/Claude rewrite
- Efficient hybrid generation: sentence-level beam search (SentBS)
- Adaptive SimMark intervals auto-tuned per text type

References

- Kirchenbauer et al. A Watermark for LLMs. ICML 2023. arXiv:2301.10226
- Aghdam et al. SimMark: Sentence-Level Watermarking. EMNLP 2025
- Pang et al. No Free Lunch in LLM Watermarking. 2024
- Liu et al. Watermark Stealing in LLMs. 2024
- Dathathri et al. SynthID-Text. Nature 2024

Website: creegon.github.io/Design_Project

Code: github.com/creegon/Design_Project